

**DESIGN AND CONSTRUCTION OF A MICROCONTROLLER-BASED
AUTOMATIC TEMPERATURE-CONTROLLED FAN WITH VOICE
COMMAND INTEGRATION**

BY

ADENIGBA OLUWAFERANMI SAMUEL

20EG02003

A PROJECT REPORT SUBMITTED TO

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING,

FACULTY OF ENGINEERING,

AJAYI CROWTHER UNIVERSITY, OYO, OYO STATE, NIGERIA

**IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE AWARD OF THE
BACHELOR DEGREE IN ELECTRICAL AND ELECTRONICS ENGINEERING.**

JUNE, 2025

CERTIFICATION

This is to certify that the project titled “Design and Construction of a Microcontroller-Based Automatic Temperature-Controlled Fan with Voice Command Integration” was carried out by ADENIGBA Oluwaferanmi Samuel with matriculation number, 20EG02003 of the Department of Electrical Electronics, Faculty of Engineering, Ajayi Crowther University, Oyo, Oyo State under the supervision of;

Prof. Z.K. Adeyemo
Project Supervisor

Date

Dr. O.A. Ajeigbe
Head of Department

Date

Prof. S.O. Jekayinfa
Dean, Faculty of Engineering

Date

ACKNOWLEDGEMENT

I give all glory and thanks to God Almighty for His unwavering guidance, strength, and support throughout the course of this project. His grace has been my anchor, and it is only by His mercy that I have come this far. I remain forever grateful.

My heartfelt appreciation goes to my supervisor, Prof. Z.K. Adeyemo, for his consistent mentorship, encouragement, and invaluable guidance, which have been instrumental to the successful completion of this work. I also extend my sincere gratitude to the Head of Department, Dr. O.A. Ajeigbe, and to all the staff of the Department of Electrical and Electronics Engineering for providing a conducive and supportive academic environment.

Special thanks to my beloved parents, Ven. Dr. Olufemi Familoni and Olori. Dr. Modupe Familoni, for their unending love, prayers, sacrifices, and encouragement. Your unwavering support at every step of this journey has been a tremendous source of strength to me. I pray that the Almighty God will continue to bless, protect, and uplift you in Jesus' name.

I would also like to express my heartfelt appreciation to my dear brother, Opemipo, and my sister, Ayomiposi, for their continuous love, support, and encouragement. Your presence in my life means more than words can express.

Lastly, I am thankful to all my friends who stood by me, inspired me, and supported me during this academic journey. Your kindness, understanding, and motivation played a valuable role in helping me stay focused and committed.

ABSTRACT

This project focuses on the Design and Construction of a Microcontroller-Based Automatic Temperature-Controlled Fan with Voice Command Integration to provide intelligent, energy-efficient cooling. Manual fan operation is often inefficient and inconvenient, especially in changing environmental conditions. While traditional thermostats and basic speed controllers have been employed to automate fan operation, they often lack precision, adaptability, and user accessibility. These existing methods do not offer real-time interactivity or hands-free control. Therefore, the aim of this project is to develop a smart, user-friendly fan system that adjusts automatically to ambient temperature and allows voice-controlled operation for added convenience and accessibility.

To achieve the project objectives, a DHT11 temperature sensor was used to continuously monitor ambient conditions, and a PIC microcontroller was programmed to adjust the fan speed accordingly via Pulse Width Modulation (PWM). A voice command module was also integrated to enable hands-free control functions like switching the fan ON/OFF or adjusting speed levels. The system was powered by a 6V rechargeable battery managed via a Battery Management System (BMS), and real-time data was displayed using an LCD. Each of these design aspects was developed in line with the project objectives: automatic fan control, voice interaction, energy efficiency, and user accessibility. The performance was then evaluated through temperature response tests, voice command response, and energy usage assessments.

During testing, the system accurately sensed temperature variations and dynamically adjusted the fan speed to match the environmental conditions. At temperatures below 26°C, the fan remained OFF. Between 26°C–31°C, it operated at speed 1 (low speed); at 32°C–37°C, it increased to speed 2 (medium speed); and from 38°C–43°C, it operated at speed 3 (high speed). As the temperature rose to 44°C–49°C, the system set the fan to speed 4 (very high speed), and for any temperature above 50°C, the fan reached speed 5 (maximum speed). Voice commands were successfully recognized for key operations, allowing users to override automatic functions or interact with the system without physical contact. The LCD displayed real-time feedback, and the rechargeable battery reliably powered the system during operation. These observations confirmed that the system met its intended objectives effectively.

In conclusion, the project successfully demonstrated a reliable and intelligent fan control system that combines automation and voice interaction. It offers practical solutions for energy conservation, ease of use, and accessibility, especially for individuals with physical limitations. It is recommended that future improvements include integration with IoT platforms for remote access and the use of higher-precision sensors to expand application scope in smart home and industrial environments.

TABLE OF CONTENT

CERTIFICATION	ii
ACKNOWLEDGEMENT	iii
ABSTRACT	iv
TABLE OF CONTENT	vi
TABLE OF FIGURES	ix
TABLE OF PLATES	x
CHAPTER ONE	1
INTRODUCTION	1
1.1 Background to the Study	1
1.2 Statement of the Problem	2
1.3 Aim and Objectives of the Study	2
1.4 Justification of the Study	3
1.5 Scope of the Project.	3
CHAPTER TWO	4
LITERATURE REVIEW	4
2.1 Room Temperature Control and Management System with Voice Command Integration	4
2.2 Automated Temperature Control and Energy-Saving Systems with Voice Commands	5
2.3 Relevant Parameters for Voice-Integrated Automated Cooling Systems	5
2.4 Outline of Automated Rechargeable Fan with Voice Command Integration	6
2.5 Automated Rechargeable Fan in Smart Home Systems	7
2.6 Reviews of Related Work	7
2.7 Research Gap	11
CHAPTER 3	12

METHODOLOGY	12
3.1 Design Approach of an Automatic Temperature Control Fan with Voice Integration	12
3.2 Construction Methodology	15
3.2.1 6V Duravolt Battery	18
3.2.2 5V Power Supply	18
3.2.3 BMS (Battery Management System)	19
3.2.4 Pic Microcontroller	19
3.2.5 LCD	20
3.2.6 DHT11 Temperature Sensor	20
3.2.8 Dc Fan	21
3.3 Mathematical Model of Automatic Room Temperature Controlled Fan with Voice Command Integration.	25
3.4 Algorithm of The Proposed System.	28
3.6 Implementation of the System	34
3.6.1 Circuit Design	34
3.6.2 Pre-Implementation Testing	34
3.6.3 Programming of the Microcontroller Unit	35
3.6.4 Hardware Implementation	35
3.6.5 Fan speed control	36
3.6.6 Soldering	36
3.6.7 Casing	37
3.6.8 Coupling	38
CHAPTER FOUR	39
RESULT AND DISCUSSION	39
4.1 Performance Test	39

4.1.1	Performance in Reading the Room Temperature	40
4.1.2	BMS (Battery Management System)	40
4.1.3	LCD Display	41
4.1.4	Speed Control	41
4.3	Presentation of Results	43
CHAPTER FIVE		53
CONCLUSION AND RECOMMENDATIONS		53
5.1	Conclusion	53
5.2	Limitations	53
5.3	Recommendations	54
REFERENCES		56

TABLE OF FIGURES

Table number		Page
3.1	Block diagram of the proposed system	14
3.2	Flow chart diagram	17
3.3	5V power supply.	18
3.4	Battery management system.	19
3.5	LCD 16X2	20
3.6	DHT11 Temperature sensor.	20
3.7	LED	21
3.8	Bluetooth module	22
3.9	Circuit diagram of automatic Temperature control rechargeable fan with voice command integration.	23
3.10	3D view of the circuit board	24
3.11	Simulation of the proposed code on proteus	33

TABLE OF PLATES

Table number		Page
3.1	6V Duravolt battery	18
3.2	Arduino Uno microcontroller.	19
3.3	12V DC fan	21
3.4	Testing using multimeter	35
3.5	16 Soldered components on board	37
3.6	Casing	37
3.7	Coupled circuit and casing.	38
4.1	Automatic temperature-controlled DC fan at speed 0	44
4.2	Automatic temperature-Controlled Fan: Speed 1 (26–31°C)	45
4.3	Automatic Temperature-Controlled Fan: Speed 2 (32–37°C)	46
4.4	Automatic Temperature-Controlled Fan: Speed 3 (38–43°C)	47
4.5	Automatic Temperature-Controlled Fan: Speed 4 (44–49°C)	48
4.6	Automatic Temperature-Controlled Fan: Speed 5 (50°C–∞)	49
4.7	Automatic Temperature-Controlled Fan: Speed 0 (No Command Detected)	51
4.8	Automatic Temperature-Controlled Fan: Speed 1 (Voice Command Activated)	51

CHAPTER ONE

INTRODUCTION

1.1 Background to the Study

The increasing demand for smart home technology has encouraged the development of gadgets that promote convenience, energy efficiency, and automation. One such gadget is the automatic temperature-controlled fan, which changes its operation based on ambient variables to maintain a comfortable climate. Traditionally, fans work with manual settings, requiring regular modification based on temperature changes. However, with improvements in microcontroller technology, it is now possible to design systems that automatically change the speed of fans according to temperature readings (Anket et al., 2021; Shivshankar et al., 2020).

A microcontroller is a compact integrated circuit designed for real-time data processing and control, widely used in automation applications due to its efficiency, affordability, and versatility. Microcontroller-based systems enable the integration of sensors, motors, and communication technologies, making them suitable for projects requiring precise monitoring and automated control, such as the autonomous fan system.

Temperature regulation is vital for enhancing comfort in both home and commercial areas. The idea of an autonomous temperature-controlled fan is to employ temperature sensors such as the LM35 or DHT11, which measure the surrounding temperature. This data is relayed to a microcontroller, which evaluates the information and adjusts the fan's speed accordingly. The fan operates at faster speeds when the temperature exceeds a specified threshold and slows down when the temperature drops, delivering an energy-efficient and responsive cooling solution. Incorporating voice command integration into such a system offers another layer of ease and user-friendliness. Modern voice recognition systems, such as Google Assistant or Amazon Alexa, can be linked with Arduino boards through platforms like ESP8266 or ESP32 for wireless control. This lets users operate the fan with simple voice requests, such as turning the fan on or off, modifying the speed, or establishing a preferred temperature range. This integration of temperature control and voice command technology gives a seamless and hands-free experience, harmonizing with the growing trend of automation in smart homes. Additionally, it illustrates the possibility of

adopting microcontroller-based solutions to solve common difficulties, increasing both comfort and efficiency in residential contexts. The development of this system also provides insight into the practical applications of IoT (Internet of Things) in enhancing everyday appliances and systems. Thus, the integration of an autonomous temperature-controlled fan with voice command technology offers a viable method to produce a more responsive, intelligent, and user-friendly home environment while also contributing to energy savings and convenience.

1.2 Statement of the Problem

In hot and humid weather, many people, especially those with physical limitations, find it difficult to manually alter fan speeds, which can be time-consuming and require continual attention. As a result, individuals often face uncomfortable room circumstances. Traditional fans in residences and workspaces also require manual adjustments to control speed, leading to inefficiencies, greater energy usage, and the hassle of frequent monitoring as temperatures fluctuate. While smart home technology has grown in popularity, many accessible temperature-controlled fans lack voice command features, which many customers prefer for hands-free operation.

To handle these difficulties, there is a need for an automated, efficient system that integrates temperature control with voice command features. Such a system would allow users to automatically change the fan's speed based on ambient temperature and give hands-free operation, boosting comfort, energy efficiency, and convenience. This project presents a Microcontroller-Based Automatic Temperature-Controlled Fan with integrated voice commands, providing a sustainable and adjustable solution that fulfills the individual needs of diverse locations while decreasing energy consumption and expenses.

1.3 Aim and Objectives of the Study

The aim of this project is to construct a Microcontroller-Based Automatic Temperature-Controlled Fan integrated with voice command functionality.

The objectives are to:

- i. develop a microcontroller-based system for automatic fan speed control based on ambient temperature.
- ii. integrate voice command functionality for hands-free fan control.
- iii. enhance energy efficiency by adjusting fan speed to maintain a comfortable temperature.

1.4 Justification of the Study

This study is essential as it addresses the growing demand for smart, energy-efficient solutions in everyday appliances. Traditional fans require constant manual adjustments, which can be inconvenient, particularly for individuals with physical disabilities. By integrating automatic temperature control and voice command functionality, this project provides an innovative solution that not only improves comfort but also reduces energy consumption.

1.5 Scope of the Project.

This project focuses on the design, programming, and testing of a Microcontroller-based automatic temperature-controlled fan coupled with voice command features. The fundamental purpose is to design a functional system where the fan's speed is automatically adjusted depending on real-time temperature measurements from a sensor, such as the LM35 or DHT11, ensuring ideal comfort without manual intervention. The project also incorporates the incorporation of voice command technologies, allowing customers to manage the fan hands-free using platforms like Google Assistant or Amazon Alexa. This ensures convenience and accessibility, particularly for persons with physical limitations. Additionally, the system will be built to optimize energy efficiency, ensuring that the fan only operates at the required speed, hence eliminating excessive power use.

CHAPTER TWO

LITERATURE REVIEW

2.1 Room Temperature Control and Management System with Voice Command Integration

A room temperature control system with voice command integration regulates and maintains the temperature of an indoor space using sensors, controllers, actuators, and advanced voice recognition technology. This system enables users to monitor and adjust temperature conditions in real-time through spoken commands, enhancing convenience and accessibility. Recent research concludes that room temperature management systems facilitate optimal comfort and energy efficiency by automating temperature adjustments (Li, Zhang, and Wang, 2018).

Such systems have become integral to modern home automation, particularly in automated cooling systems. The integration of microcontrollers, temperature sensors, and voice command interfaces forms a robust solution for maintaining thermal comfort. These systems are highly effective for residential, commercial, and institutional buildings, providing energy-efficient and user-friendly operation (International Energy Agency, 2019). Over time, there have been multiple investigations carried out to assess the precision and dependability of temperature management systems, here are some of the important findings from the research:

- i. Temperature Management

Room temperature control systems, traditionally dependent on manual adjustments or basic programmable thermostats, have evolved with technological advancements. Systems now employ microcontrollers, sensors, and voice interfaces for real-time automatic adjustments based on temperature readings. For instance, Arduino microcontrollers offer affordability and flexibility for developing such systems (Dinh, Nguyen, and Le, 2019).

- ii. Temperature Sensing

Temperature sensors, such as the DHT11, are critical components that provide precise real-time data for automated control. When combined with voice recognition, users can issue commands to set desired temperature levels, enhancing interactivity and ease of use (Chou, Chen, and Hsu, 2014).

2.2 Automated Temperature Control and Energy-Saving Systems with Voice Commands

Modern automated cooling systems emphasize energy efficiency, precision, and ease of operation. Voice command integration further improves usability by enabling hands-free control of fan and temperature settings.

Key Features:

i. Control and Automation

Unlike traditional fans, modern automated fans respond dynamically to temperature changes, user preferences, and occupancy. Voice command functionality allows users to adjust fan speed or turn the fan on and off without manual input, enhancing flexibility and convenience (Wang, Zhang, Li, and Chen, 2021).

ii. Energy Efficiency

Voice-integrated automated fans optimize energy usage by incorporating features like occupancy monitoring and scheduling. Users can issue commands such as “reduce fan speed” or “turn off after 30 minutes,” improving energy conservation (Saksena, Verma, and Mehta, 2021).

iii. Precision and Customization

Voice commands enable precise control of fan settings, such as speed and oscillation. Users can specify commands like “set speed to medium” or “turn on cooling mode,” providing a personalized experience (Kose, Kaya, and Aydin, 2019).

2.3 Relevant Parameters for Voice-Integrated Automated Cooling Systems

The following parameters are essential for the design and functionality of a voice-integrated automated cooling system:

i. Temperature

Temperature sensors provide data for controlling fan speed. Voice commands such as "set temperature to 25°C" allow users to directly control the system's response to environmental conditions (Ouyang, Lin, and He, 2021).

ii. Humidity

Humidity sensors inform decisions to adjust fan speed or activate specific modes. Voice commands like “reduce humidity” can trigger optimized operations (Tham, Cheong, and Sekhar, 2015).

iii. External Conditions

Sensors monitor ambient conditions, enabling users to give commands such as “adjust to outdoor mode” for energy-efficient operation during favourable weather (Jo, Park, and Kim, 2018).

iv. Speed Control

Variable speed control ensures precise adjustment to airflow needs. Voice commands like “increase speed” or “switch to silent mode” enhance user experience (Garg, Nair, and Sharma, 2020).

v. Energy Monitoring

Users can monitor energy consumption and issue commands such as “check power usage” to optimize the system's efficiency (Subramanian, Ramesh, and Iyer, 2019)

2.4 Outline of Automated Rechargeable Fan with Voice Command Integration

Automated rechargeable fans with voice command integration combine portability, intelligence, and user-friendliness. They feature advanced algorithms, sensors, and rechargeable batteries to provide adaptive cooling based on environmental conditions and user preferences.

Construction Steps:

- i. **Component Selection:** Choose microcontrollers, temperature sensors, voice modules, and DC motors.
- ii. **Integration:** Connect the temperature sensor and voice module to the microcontroller.
- iii. **Programming:** Develop software to adjust fan speed based on temperature readings and execute voice commands.
- iv. **Battery and Charging System:** Incorporate a rechargeable battery with charging circuitry.
- v. **Testing:** Validate the system’s functionality and ensure safety standards.

These fans automatically adjust their speed and airflow in response to spoken commands such as “increase fan speed” or “set temperature to 22°C,” making them suitable for diverse applications (Hoque, Rahman, and Ahmed, 2019).

2.5 Automated Rechargeable Fan in Smart Home Systems

Integrating voice-controlled rechargeable fans into smart home ecosystems enhances functionality and user convenience. These fans can be controlled via voice commands or mobile apps, offering features like energy-efficient operation, remote accessibility, and coordination with other smart devices.

Key Benefits:

- i. Voice Command Control: Enables hands-free operation with commands like “turn on fan” or “switch to night mode.”
- ii. Energy Efficiency: Intelligent control algorithms and DC motors reduce power consumption.
- iii. Air Quality Improvement: Integration with air quality sensors allows voice commands like “purify air” to trigger appropriate responses (Rathi, Deshmukh, & Bhagat, 2020).

Voice integration transforms automated fans into versatile tools for enhancing indoor comfort, ensuring energy efficiency, and contributing to smart home automation.

2.6 Reviews of Related Work

This subsection discusses the review of works that has been done by other researchers with regards to Temperature controlled fan and voice command integration.

Nur Afqah Junizan. (2019) Design and Implementation of Automatic Room Temperature Controlled Fan using Arduino Uno and LM35 Heat Sensor. This project solves the inconvenience of manually adjusting fan speed as room temperatures fluctuate by automating the process. The system makes living more comfortable and energy-efficient without needing any effort from the user. It employs an LM35 temperature sensor to keep track of the room’s temperature and an Arduino Uno microcontroller to process the data. By employing pulse-width modulation (PWM) technology, the system adjusts the fan’s speed dependent on how hot or chilly the room gets. When the temperature passes a predetermined threshold, the fan automatically comes on, and its speed

increases as the temperature rises. An LED lights up to show when the fan is engaged, and the fan continues until the temperature goes below the threshold.

The results reveal that the system works efficiently, monitoring temperature in real-time, regulating fan speed automatically, and maintaining a comfortable environment. The PWM technology ensures smooth, exact speed changes, saving energy while eliminating the need to regulate the fan manually. The system's key features are its ease, energy economy, and possibility for use in diverse domains, such as HVAC systems or industrial cooling. That so, it has a few drawbacks, such as a lack of options for changing temperature thresholds or speed levels, possible reliability concerns with the sensor, and no consideration for other elements like humidity. Despite these shortcomings, this project is a realistic and efficient solution for automatic fan control, making it a great step forward in temperature management.

Jian Rong Fan. (2019) Energy Saving - Motion Activated Smart Fan Design and Implementation. This project addresses the environmental impact of air conditioning systems, which release harmful CFCs, and the inefficiency of standard fans that run nonstop, whether or not someone is present. The smart fan offers a practical, eco-friendly solution by adapting its operation to user presence and movement, using an IR sensor to detect when a user is nearby and turn the fan on or off automatically, and an ultrasonic sensor to track the user's movements and focus airflow where it's needed most. These features ensure the fan only runs when necessary, saving energy and providing targeted cooling for maximum comfort. Results show that the smart fan reduces power consumption by 11.47% compared to standard fans, offering an efficient and thoughtful alternative to traditional cooling methods. Its strengths include reduced environmental impact, energy efficiency, and user-focused design, though it has some limitations, such as limited suitability for multiple users, potential sensor accuracy issues, and higher complexity and cost. Overall, the smart fan is a sustainable and innovative solution that balances energy savings and user comfort, aligning closely with my work on automatic fan control systems, which focus on temperature-based fan speed adjustments. Combining user presence and motion detection with temperature control could lead to an even more advanced and versatile system.

Amiri Nazanin. (2007) Voice-Activated Remote Control for Fan/Heater. This project solves the problems with regular remote keys by creating a voice-controlled alternative. This makes it easier for everyone to use, especially older or disabled people. The system uses a microcontroller and a

voice recognition IC (HM2007) to process voice commands and trigger the related actions, such as pressing buttons on the remote control. The project successfully shows that voice-activated remote controls are possible, which will cut down on manual work and make users more comfortable. But the system might not work well in noisy places because the voice recognition chip isn't very good at that, and it needs pre-recorded orders to work. Despite these limits, the project shows the potential of voice recognition technology for improving the accessibility and usability of electrical appliances. This aligns with my work on the "Design and Implementation of a Temperature Controlled Fan with Voice Command Integration", as both projects combine voice control with microcontroller technology to improve user experience.

Khan, Hashim Raza. (2024) Design and Experimental Results of an AIoT-Enabled, Energy-Efficient Ceiling Fan System. This research addresses the inefficiency of ceiling fans in energy consumption by creating an IoT-based system for optimizing their operation. The methodology involved retrofitting standard DC fans with an IoT circuit using an ESP8266 microcontroller, temperature, humidity, and motion sensors. Data was collected over two years, first with manual operation and then with an AI-based auto mode. Results showed a 50% reduction in energy loss compared to traditional AC fans, significantly lowering carbon footprints and receiving high user acceptability. The system's strengths include improved energy efficiency, reduced environmental impact, and user satisfaction. However, it relies on consistent IoT connectivity and has a higher upfront cost. This project relates to my work on "Design and Implementation of a Temperature Controlled Fan with Voice Command Integration" by focusing on improving fan functionality and efficiency through advanced technologies like microcontrollers and sensors.

Da Silveira (2023) Thermal and wind devices for multisensory human-computer interaction. This paper explores the integration of thermal and wind effects into human-computer interaction (HCI) to enhance user experiences. While current systems can generate airflow, they struggle to combine wind with temperature changes, such as producing hot or cold air. This is due to the complexity of integrating blowing mechanisms with heating or cooling effects, as airflow often disrupts the temperature control. The study reviews wearable and non-wearable devices, analysing their features and limitations based on user feedback. Wearable devices provide localized effects but can be uncomfortable, while non-wearable systems are less intrusive but lack precision and are influenced by environmental factors. A review of commercial products reveals a shift toward IoT-

based systems, offering potential for more immersive technologies. The findings highlight that while wind devices are effective for airflow, achieving combined thermal and wind sensations remains a challenge, presenting opportunities for innovation. This research connects to my project, "Design and Implementation of a Temperature Controlled Fan with Voice Command Integration," as both focus on enhancing user comfort by combining airflow, temperature control, and smart technologies for efficient, user-centred solutions.

Fu Yang (2010) Design an Automatic Temperature Control System for Smart Tudung Saji Using Arduino Microcontroller. By combining hardware and software into an automatic food cover, the Smart Tudung Saji project makes it possible to keep food warm while keeping pests like flies and rats away. It uses an Arduino Uno microprocessor, an LM35 temperature sensor to check the temperature of the food, and a limit switch sensor to find out where the lid is. Two 240V 100W bulbs are used as heating elements in the system. They turn on when the temperature goes below 60°C and turn off automatically when it reaches 60°C, making sure that the room is properly warmed. Simulations in Proteus 6 Lite and tests with the prototype proved that the design was good enough to be used every day, especially by people who are working. The project is cheap and easy to move around, but it has some problems, like using a lot of energy and being limited to a certain temperature range. It might use less energy if the lights were changed out for thermistor coils. This project is very similar to mine because it also uses automation and temperature control. However, my project is more focused on cooling and adds voice prompts to make it more useful.

Wen, Y., and Jeng, M. (2008) Diagnosable discrete event systems design. In this paper, the tricky problem of diagnosability in discrete event systems is looked at in depth. This is especially important in high-tech fields like making semiconductors. The writers show a new way to describe and study how systems work by using Petri nets. They then turn this into a binary integer linear programming (BILP) problem. This method not only finds problems with diagnosability, but it also offers smart places to put sensors to help find and isolate faults. One great thing about this method is that it can be used in real life. The writers put their method to the test on a real Metal-Organic Vapor Phase Epitaxy (MOVPE) system and showed that it can successfully separate event cycles that are not different from each other. It's important to keep in mind, though, that this method is based on the idea that sensing costs are very low, which may not always be the case. Even though it has some flaws, this work is very helpful because it shows how

Petri nets and BILP can be used in systematic analysis. Its focus on sensor-driven automation fits well with my project, which is all about controlling temperature and making things easier for users by letting them use voice requests. Our projects are different in what they cover, but we all want to achieve the same thing: to improve system performance through structured methods and sensor integration.

2.7 Research Gap

While several studies have explored temperature-controlled fan systems and others have examined voice command integration separately, very few have combined both technologies into a single, low-cost and microcontroller-based design. Most existing solutions either lack real-time adaptability to temperature changes or do not provide voice control features for enhanced user convenience. This project addresses that gap by designing and constructing an integrated system that offers automatic temperature-based fan speed control along with voice command functionality, improving both energy efficiency and accessibility.

CHAPTER 3

METHODOLOGY

3.1 Design Approach of an Automatic Temperature Control Fan with Voice Integration

The entire system of the design of an automatic temperature-control rechargeable DC fan with voice command integration will be made up of a pic microcontroller, DHT11 temperature sensor, LCD display, a 5V power supply, Voice command module, rechargeable battery, DC fan, and the control unit. DHT11 Temperature Sensor measures the ambient temperature and provides digital output to the pic microcontroller. The Pic microcontroller reads the temperature readings from the DHT11 temperature sensor and makes decisions based on the temperature. It also controls the LED and the DC fan accordingly and displays the current temperature on the LCD display. The battery serves as a portable power source for the system which is connected to the battery management system. The battery management system ensures safe charging and discharging of the battery, which is connected to the 5V power supply. The 5V Power supply provides power to the Pic microcontroller and other components. The LED serves as an indicator to show the system's status, such as whether the fan is running or not. The Voice command module allows the fan to be controlled wirelessly with voice command integration. The DC fan, which is the primary output component that adjusts its speed based on temperature readings to regulate the temperature. The block diagram of the proposed system provides a comprehensive overview of the system's architecture and interconnection, offering insights into its potential practicality in real-world settings.

The following are components in the constructional set up:

- i. 22pf ceramic capacitor - 2x
- ii. 1000uf/25v electrolytic capacitors
- iii. 100uf/16v electrolytic capacitor
- iv. 8mhz crystal oscillatory capacitor
- v. 10k ohms resistor - 5x
- vi. 1k ohms resistor - 4x
- vii. 2.2k ohms resistor

- viii. 100 ohms resistor
- ix. 102 variable resistor
- x. pic16f876a microcontroller
- xi. 28pin dip ic sit
- xii. dht11
- xiii. 7805
- xiv. bc547 npn transistor - 2x
- xv. 1n4148 signal diode
- xvi. 1n4007 rectifier diode
- xvii. light emitting diode
- xviii. hc05 bluetooth module
- xix. 16x2 lcd display
- xx. 5v/10A relay
- xxi. power switch
- xxii. press-to-make button - 3x
- xxiii. 2pin terminal block - 2x
- xxiv. jumper connectors
- xxv. junmper wires
- xxvi. pcb copper plate
- xxvii. pcb photopaper
- xxviii. soldering lead

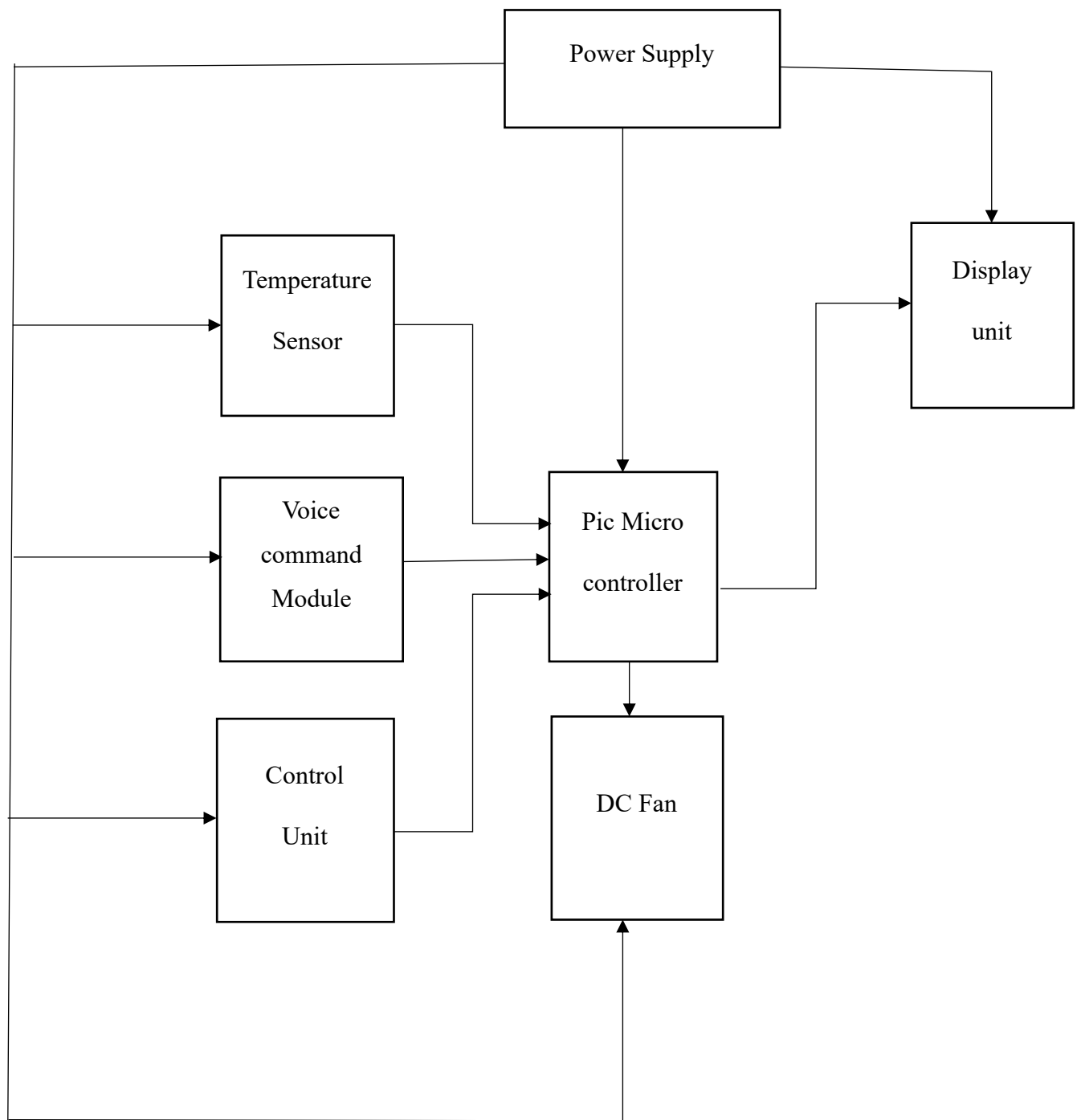


Figure 3.1 Block diagram of the proposed system

3.2 Construction Methodology

The DC fan, control unit, LCD display, rechargeable battery, 5V power source, and PIC microcontroller are all part of the temperature-regulating fan system. The system is designed to be user-friendly, with voice command integration allowing for hands-free control. The basic idea is to take the temperature reading from the DHT11 sensor and alter the fan speed accordingly. When the temperature rises over a certain limit, the system is programmed to activate the fan and ramp up the speed until the maximum speed is reached in direct proportion to the temperature of the room. Through voice commands, the user can control the fan, adjust the set temperature threshold, or even turn off the system entirely.

At the heart of the system, the LCD display is initialized, and the input/output pins for the fan, LED, and DHT11 temperature sensor are configured in the setup procedure. A 6V Duravolt rechargeable battery is used as the primary power source and is first connected to a Battery Management System (BMS) for safe charging and discharging. From the BMS output, the power is split into two lines: one line supplies the 6V DC fan directly, while the second line is routed through a 5V voltage regulator. This regulator steps down the voltage to 5V, which is then used to power the PIC microcontroller, DHT11 sensor, LCD display, LED indicator, and the voice command module. The positive red wire of the DC fan connects directly to the 6V output from the BMS, and the negative black wire is linked to ground, with its speed controlled by a transistor via a PWM signal from the microcontroller. A delay of one millisecond is included in the setup to allow the system to stabilize before beginning regular operations.

The DHT11 sensor's temperature is retrieved using the `readTemp` function in the `loop` function. Alongside this, a voice recognition module listens for specific voice commands such as "Turn on the fan," "Set temperature to 28 degrees," or "Turn off overheating alerts." When a voice command is received, the system processes it and either overrides the automatic temperature control or adjusts the fan's behaviour as per the user's instructions. For example, if the user commands, "Increase fan speed," the system will respond immediately by increasing the fan's speed, regardless of the temperature reading.

When the temperature drops below the minimum set point, both the fan speed and the fan LCD variable are set to 0. If the temperature is within the range of the lowest and maximum temperatures, the fan speed is calculated based on the measured temperature, and the fan LCD

variable is updated to reflect the percentage of fan speed. The system sends an analogWrite signal to the fan pin to achieve the desired speed. If the temperature rises above the threshold, the LED lights up to warn of overheating. Voice commands such as "Disable fan" or "Set threshold to 35 degrees" can dynamically control these behaviours, making the system adaptable to user needs.

The LCD display plays a vital role in providing real-time feedback. Using the LCD. Print function, the temperature and fan speed are displayed on the screen. A delay of 200 ms is added to prevent flickering, and the LCD. Clear function is used to clear the screen before the next cycle begins. Voice commands also influence the display; for instance, if the user says, "Show current settings," the system will display the target temperature and fan status for user convenience.

The readTemp function reads the DHT11 sensor's temperature and converts it to degrees Celsius. To get the temperature in Celsius, the reading is multiplied by 0.48828125. The closed-loop fan system for temperature regulation constantly compares the current and target temperatures. If the real temperature deviates from the setpoint, the fan speed is adjusted, and the LCD screen updates accordingly. The voice command integration adds a new layer of interactivity, allowing users to control the system seamlessly, without needing to interact with physical buttons or switches.

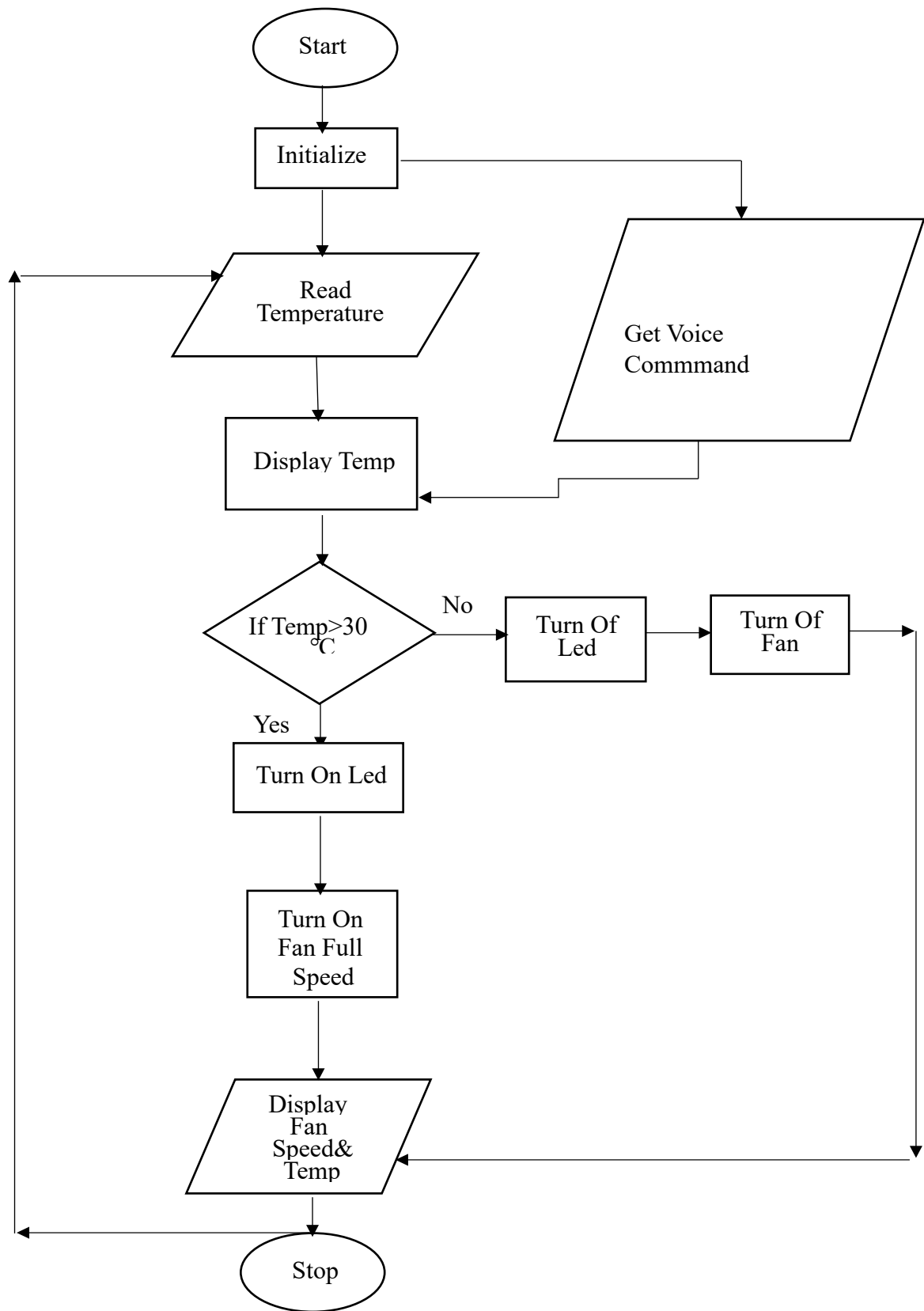


Figure 3.1 Flow chart diagram

3.2.1 6V Duravolt Battery

The 6v duravolt battery is a type of rechargeable battery that supplies direct power to the fan and indirect power to other components through a voltage regulator.



Plate 3.1 6V Duravolt battery

3.2.2 5V Power Supply

A 5V power supply is a device or circuit that provides a steady voltage output of 6V. It can be used to power electronic devices that require a constant voltage to operate properly. These power supplies can be found in many different forms, including wall adapters, USB chargers, and internal power supplies for computers and other electronic devices.



Figure 3.3 5V power supply.

3.2.3 BMS (Battery Management System)

The process of charging and discharging a battery is managed by a battery management system (BMS). It prevents the battery from being damaged by excessive charging, discharging, or heating, and guarantees a safe and effective charging and draining process.



Figure 3.4 Battery management system.

3.2.4 Pic Microcontroller

It serves as the core processing unit for the automatic temperature-controlled fan with voice command integration handling key functions such as; temperature monitoring and speed control, voice command processing, power management, display and feedback system, communication and user interface.



Plate 3.2 Arduino Uno microcontroller.

3.2.5 LCD

A liquid crystal display (LCD) screen is a type of electronic display module. LCD 162 refers to a display with 16 characters across 2 lines. This LCD uses a 5x7 pixel matrix for each letter, which represents the speed of the fan.



Figure 3.5 LCD 16X2

3.2.6 DHT11 Temperature Sensor

The DHT11 temperature sensor has an accuracy of 2 degrees Celsius and can measure temperatures from 0 to 50 degrees Celsius. It has a relative humidity measuring range of 20%90%, with an accuracy of 5%. The low power consumption of the sensor makes it ideal for use in portable, battery-operated devices.

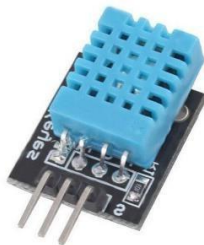


Figure 3.6 DHT11 Temperature sensor.

3.2.7 LED

When an electric current is sent through a semiconductor device known as an LED (Light Emitting Diode), light is emitted. LEDs are utilized in everything from electrical devices to car headlights to traffic signals.



Figure 3.7 LED

3.2.8 Dc Fan

A 12V portable DC fan is a compact and versatile cooling device that operates on a 12-volt power supply. It is lightweight, portable, and suitable for personal cooling, camping, or providing airflow in confined spaces. These fans offer voltage compatibility, energy efficiency, quiet operation, variable speed control, and various mounting options.



Plate 2.3 12V DC fan

3.2.9 Voice Command Module

The Voice Command Module is a hardware component that enables voice-based control of the system. It incorporates an HC-05 Bluetooth module, which facilitates wireless communication with the microcontroller by transmitting digitally decoded voice command signals for processing and execution.

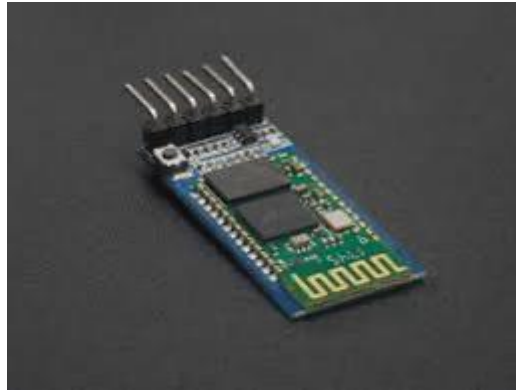


Figure 3.8 Bluetooth module

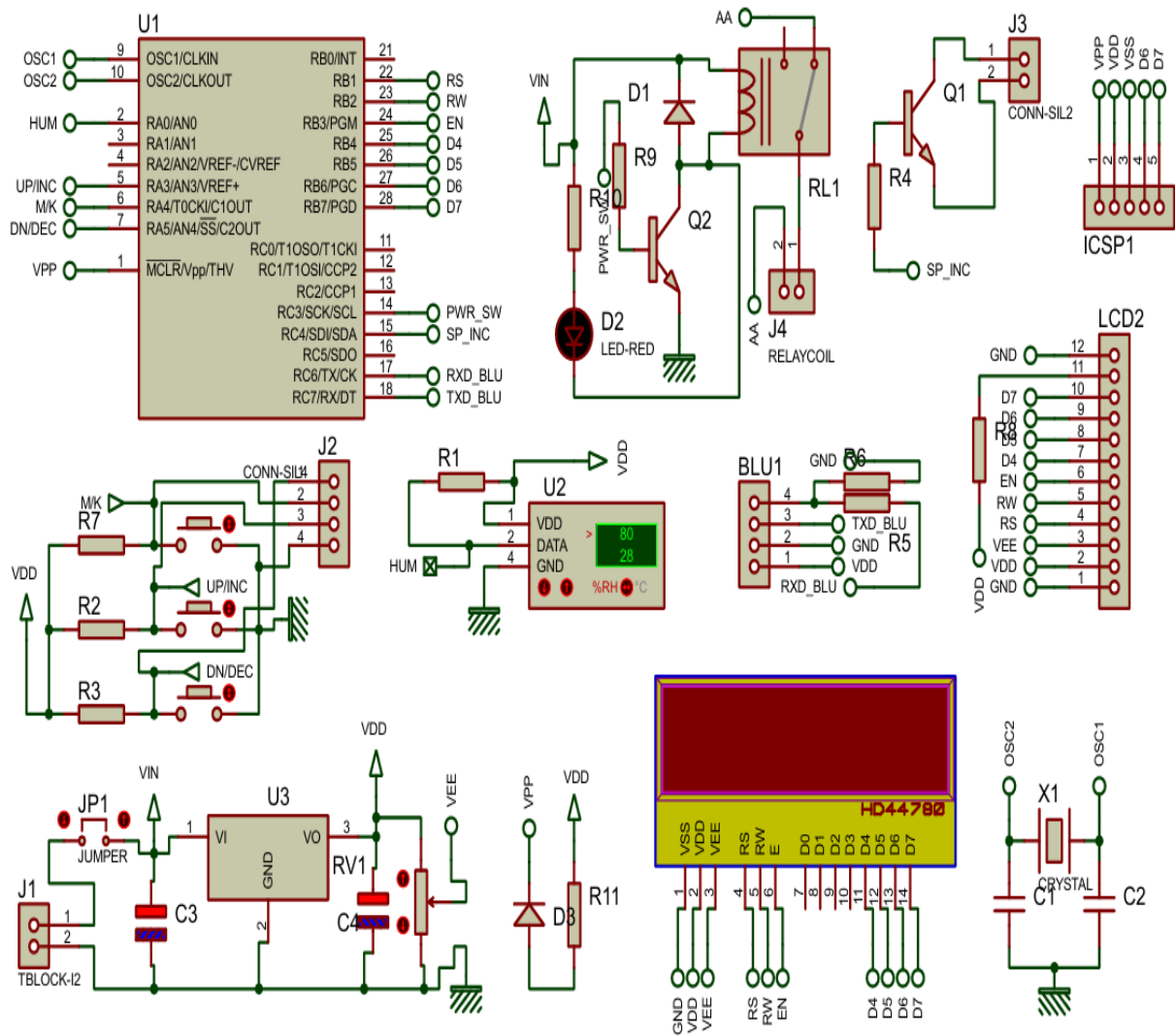


Figure 3.9 Circuit diagram of automatic Temperature control rechargeable fan with voice command integration.

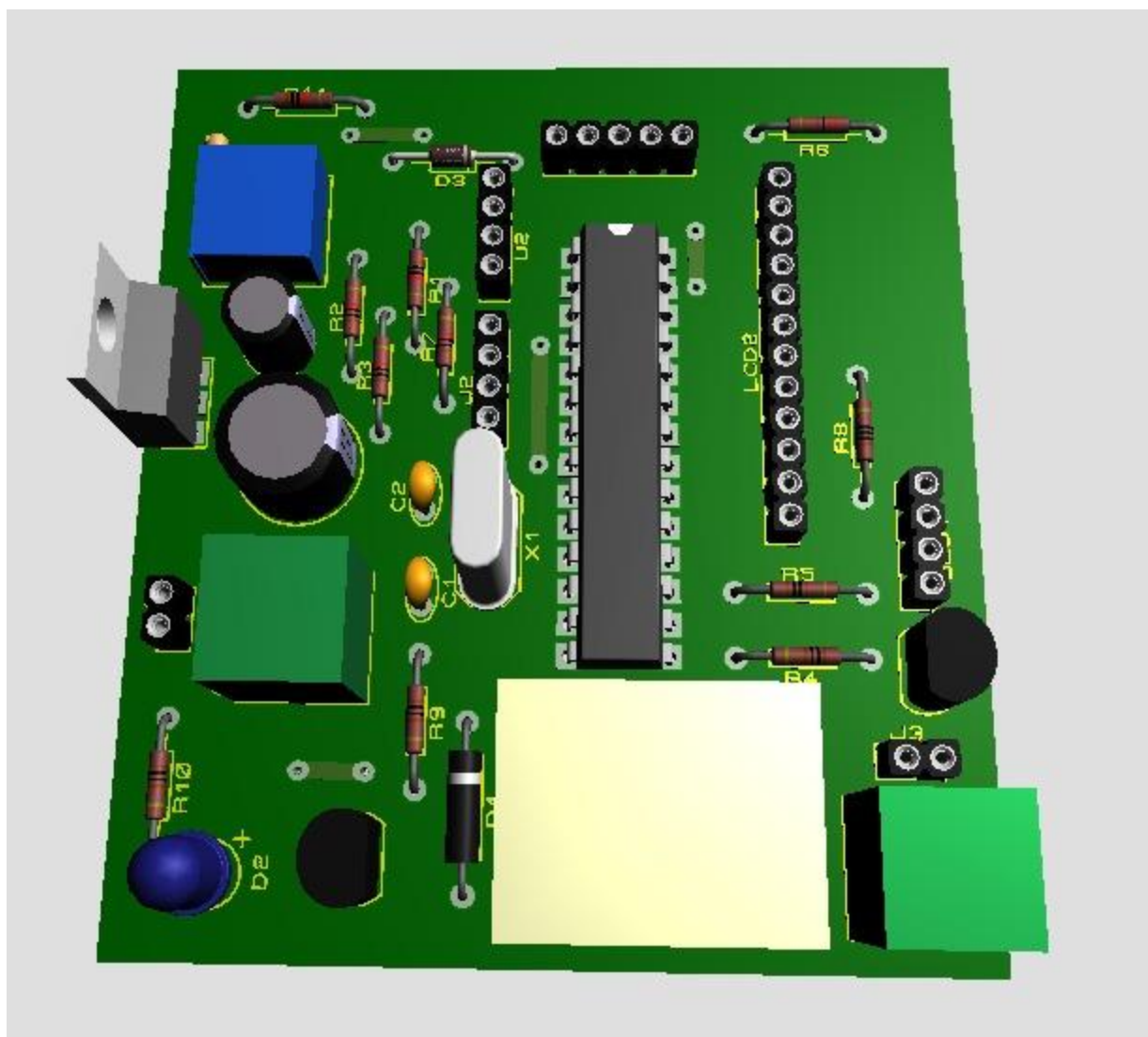


Figure 3.10 3D view of the circuit board

3.3 Mathematical Model of Automatic Room Temperature Controlled Fan with Voice Command Integration.

The mathematical model for an automatic room temperature control system with voice command integration can be broken down into several key components, with a primary focus on the fan control mechanism, temperature regulation, and the integration of voice commands. Below is an overview of how each aspect can be modelled mathematically.

1. Temperature-Based Fan Speed Control

The fan speed is automatically adjusted based on the room temperature, measured by a DHT11 sensor.

Definitions:

- i. T_{actual} = Current room temperature
- ii. T_{set} = Temperature set by the user (via voice command)
- iii. S_{fan} = Fan speed (0% - 100%)
- iv. T_{min} = Minimum temperature threshold (fan OFF)
- v. T_{max} = Maximum temperature threshold (fan at full speed)

Fan Speed Equation:

$$S_{fan} = \begin{cases} 0, & \text{if } T_{current} \leq T_{min} \\ \frac{T_{current} - T_{min}}{T_{max} - T_{min}} \times 100 & \text{if } T_{min} < T_{current} < T_{max} \\ 100, & \text{if } T_{current} \geq T_{max} \end{cases}$$

How It Works:

- i. If $T_{actual} \leq T_{min}$, the fan is OFF.
- ii. If $T_{actual} \geq T_{max}$, the fan runs at 100% speed.
- iii. If $T_{min} < T_{actual} < T_{max}$, the fan speed increases linearly with temperature.

2. Voice Command Integration

Voice commands allow users to override the temperature-based fan control.

Voice Control Actions:

- i. Turn fan ON/OFF
- ii. Set a new temperature threshold
- iii. Manually adjust fan speed

Fan Speed Adjustment via Voice Command:

$$S_{fan} = \begin{cases} 0, \\ 100, \\ S_{com}, \end{cases}$$

if user says "Turn off fan"

if user says "Turn on fan"

if user sets a specific speed

3. Power Consumption Model

Since the fan runs on a battery, power consumption is an important factor.

Power Consumption Formula:

$$P_{fan} = P_{max} \times \frac{S_{fan}}{100}$$

Where:

- i. P_{max} = Maximum power (e.g., 5W)
- ii. P_{fan} = Power consumed by the fan at any given speed

Battery Energy Equation:

$$E_{battery}(t) = E_{battery}(t-1) - P_{fan} \times \Delta t$$

This shows that the battery's energy decreases over time based on the fan's power usage.

4. Final System Control Model

$$S_{fan} = \begin{cases} \frac{T_{current} - T_{min}}{T_{max} - T_{min}} \\ 0, \\ 100, \\ S_{command}, \end{cases}$$

if no voice command and $T_{min} < T_{current} < T_{max}$

if no voice command and $T_{current} \leq T_{min}$

if no voice command and $T_{current} \geq T_{max}$

if user sets a specific speed via voice command

5. Example Scenario

Given:

- $T_{min} = 22^{\circ}\text{C}$, $T_{max} = 30^{\circ}\text{C}$
- $T_{actual} = 25^{\circ}\text{C}$
- User says: "Set fan speed to 60%"

Result:

- The fan overrides automatic control and runs at 60% speed.
- If no voice command is given:

$$S_{fan} = \frac{(25 - 22)}{(30 - 22)} \times 100 = 37.5\%$$

3.4 Algorithm of The Proposed System.

Step 1: Initialize the Microcontroller, LCD display, DHT11 temperature sensor, LED, and DC fan.

Step 2: Set the pin modes for the LED, DC fan, and LCD display.

Step 3: Set the temperature threshold value (30-50°C) for fan activation.

Step 4: Initialize variables to store temperature readings.

Step 5: Read temperature from the DHT11 sensor.

Step 6: Display the temperature on the LCD display.

Step 7: Check for incoming voice commands via Bluetooth.

Step 8: If a voice command is received: process the voice command using a voice recognition algorithm.

Step 9: If the voice command is to turn on/off the fan: update the fan state accordingly.

Step 10: If the voice command is to set a new temperature threshold: update the temperature threshold value.

Step 11: Update the LCD display to reflect the new temperature threshold

Step 12: Check if the temperature exceeds the threshold value.

Step 13: If the temperature is higher than the threshold, turn on the LED, start the DC fan at full speed and update the LCD display to indicate that the fan is running.

Step 14: If the temperature is within the desired range, turn off the LED, Stop the DC fan and Update the LCD display to indicate that the fan is not running.

Step 15: Wait for a specified interval (e.g., 1 second) before repeating the process.

Step 16: Go back to step 5 and continue monitoring the temperature.

Step 12: End the algorithm.

3.5 System Code of the Proposed System.

```
#include <stdio.h>

#include <stdlib.h>

#include <string.h>

// Pin Definitions

#define DHT11_PIN PIN_A0

#define FAN_PWR_SW PIN_C3

#define FAN_SPD_SW PIN_C4

#define UP_INC PIN_A3

#define MENU_OK PIN_A4

#define DOWN_DEC PIN_A5

// Global Variables

unsigned int8 min_therm = 30, max_therm = 40, fan_speed = 0, opr_mode = 0;

float temp_val = 0;

// Function Prototypes

void sys_init(), read_dht11_temp(), menu_routine();

void set_fan_speed(unsigned int8 speed);

void decode_voice_command(char *command);

// Main Loop

void main() {

    sys_init(); // Initialize system

    while (1) {

        read_dht11_temp();
```

```

if (opr_mode == 0) { // Auto Temperature Control

    fan_speed = (temp_val < min_therm) ? 0 :

        (temp_val > max_therm) ? 5 :

            ((temp_val - min_therm) * 5) / (max_therm - min_therm);

    }

}

else if (voice_command_received) {

    decode_voice_command(voice_command);

    voice_command_received = 0;

}

}

set_fan_speed(fan_speed);

if (input(MENU_OK)) menu_routine();

}

}

}

// System Initialization

void sys_init() {

    min_therm = read_eeprom(1);

    max_therm = read_eeprom(2);

    opr_mode = read_eeprom(3);

    fan_speed = read_eeprom(4);

}

```



```

// Read Temperature from DHT11

void read_dht11_temp() {

    temp_val = 25.5; // Placeholder, replace with actual sensor reading

}

// Set Fan Speed (0-5)

void set_fan_speed(unsigned int8 speed) {

    write_eeprom(4, speed);

}

// Decode Voice Commands

void decode_voice_command(char *command) {

    char *levels[] = {"ZERO", "ONE", "TWO", "THREE", "FOUR", "FIVE"};

    for (int i = 0; i < 6; i++) {

        if (strstr(command, levels[i])) {

            set_fan_speed(i);

            break;

        }

    }

}

}

// Handle Menu Settings

void menu_routine() {

    write_eeprom(1, min_therm);

    write_eeprom(2, max_therm);

```

```
    write_eeprom(3, opr_mode);  
}
```

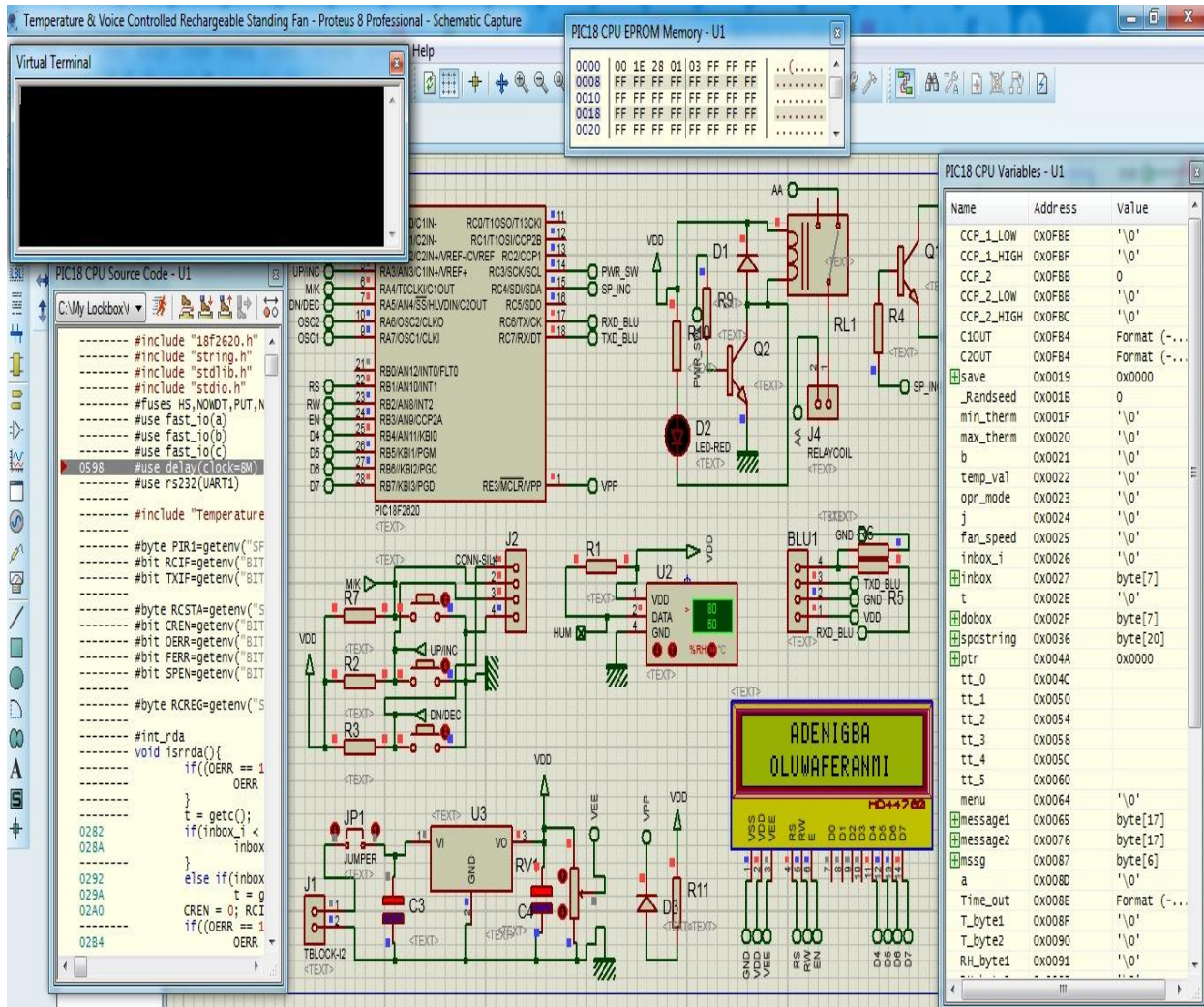


Figure 3.11 Simulation of the proposed code on proteus

3.6 Implementation of the System

In implementing any electronic circuit, there are some step-to-step procedures which are needed to be put into consideration before achieving the desired result. Steps taken during the design of the project includes:

- i. Circuit design
- ii. Pre-implementation testing of components part
- iii. Programming of the system
- iv. Hardware implementation of circuit design
- v. Fan speed control
- vi. Soldering
- vii. Casing Coupling of the circuit

3.6.1 Circuit Design

The circuit was designed on Proteus by selecting the required components from its database, which were then interconnected to form the basis of the circuit diagram as shown in Figure 3.9. Afterward, the design code was uploaded using a PIC microcontroller, and the design was executed.

3.6.2 Pre-Implementation Testing

The components were tested with a multimeter at key points throughout the project to ensure they were functioning correctly. During the assembly process, the multimeter was used to check continuity, resistance, voltage, current, and the operational status of components such as transistors and diodes. Additionally, the multimeter was used to verify the various output voltages of the power supply. Components like the battery management system, liquid crystal display (LCD), digital thermometer (DHT11), direct current (DC) fan, control board, and PIC microcontroller unit were all thoroughly tested to ensure proper functionality.



Plate 3.4 Testing using multimeter

3.6.3 Programming of the Microcontroller Unit

The microcontroller requires a program to perform the desired operation. In this case, the source code was written to instruct the PIC microcontroller to send signals to the transistor, controlling the ON and OFF states of the fan based on the temperature readings from the DHT11 sensor. Additionally, the system integrates voice command functionality to control the fan. The voice commands are processed by the microcontroller, allowing the user to turn the fan on or off via voice input. The source code, written in C-programming language compatible with the PIC microcontroller, was uploaded onto the device using Proteus software. This programming process enables the microcontroller to execute the required actions based on the temperature sensor's input or the voice command, turning the fan on or off as necessary.

3.6.4 Hardware Implementation

In this project, before the components were permanently soldered, they were first arranged on a breadboard. This stage served as the prototype phase, allowing us to verify that the circuit was functioning as expected. The temperature control fan regulator was divided into two main function blocks: the breadboard connections and the simulation on Proteus. Each block was tested and

confirmed to work according to specification. Once both blocks were validated, the final construction was carried out in an adaptable box, following the designed circuit as illustrated in Figure 3.9 above.

3.6.5 Fan speed control

In the temperature-controlled fan project, the fan speed is regulated using PWM (Pulse Width Modulation) signals. The PWM value is determined based on the temperature readings from the sensor. A scaling factor is used to convert the temperature into a corresponding PWM signal that controls the fan speed. As the temperature increases, the fan speed should also increase to maintain an appropriate temperature range or ensure effective cooling. The scaling factor adjusts the fan speed response to temperature changes, and it can be modified depending on the desired fan behaviour. The calculated PWM value should be limited to the range supported by the fan to avoid over-driving the motor or producing undesired results.

3.6.6 Soldering

After the positioning of the components on the board as provided on the circuit diagram of the project, the joining of components to one another was done using a soldering iron and a soldering lead, most components are very sensitive to heat, so extra care was taken in handling and soldering to avoid burning of the components. Also, each joint was ensured to be electrically sound to allow conduction of electricity and mechanically sound to avoid getting disjointed on receiving considerable mechanical stress.



Plate 3.5 Soldered components on board

3.6.7 Casing

The casing was constructed putting the size of the circuit board. Power supply and ventilation for the circuit into consideration.



Plate 3.6 Casing

3.6.8 Coupling

The circuit was attached to the pole by screwing it tightly to it. This is to makes it more user friendly and easily accessible.



Plate 3.7 Coupled circuit and casing.

CHAPTER FOUR

RESULT AND DISCUSSION

4.1 Performance Test

The results of testing the automatic temperature-controlled rechargeable DC fan are presented in this section. The system was thoroughly evaluated to determine its effectiveness and reliability in real-world conditions. The evaluation focused on several key performance criteria, including accurate temperature sensing, responsive fan speed control, efficient power management, and seamless integration of voice command functionality.

The temperature readings were taken using the DHT11 sensor, and the measured values were successfully displayed in real time on the LCD screen, providing the user with clear and continuous feedback on the ambient room temperature. As the temperature varied, the system automatically adjusted the speed of the DC fan using PWM (Pulse Width Modulation) signals generated by the microcontroller. This ensured that higher temperatures resulted in higher fan speeds, thereby improving cooling efficiency while maintaining energy conservation at lower temperatures.

In addition to automatic control, the system incorporated voice command capabilities to enhance user interaction and convenience. Through simple voice inputs, users were able to turn the fan ON or OFF and increase or decrease the fan speed manually, without physical contact. This feature adds a smart and modern touch to the system, making it especially useful in scenarios where hands-free operation is preferred, such as for elderly or physically challenged users.

The rechargeable battery system was also tested to ensure it could supply stable power to the fan and control system during periods when mains power was unavailable. Battery performance, voltage stability, and overall power efficiency were monitored and found to be satisfactory for the intended application.

Overall, the system demonstrated a high level of functionality and reliability across all tested areas. The integration of automatic control with voice command input proved to be both practical and efficient, enhancing the overall user experience. These results validate the success of the design and confirm that the system can serve as a viable solution for intelligent and energy-efficient temperature-based fan control.

4.1.1 Performance in Reading the Room Temperature

The system accurately measures the ambient room temperature using the DHT11 temperature sensor integrated into the design. This sensor is responsible for continuously monitoring the surrounding environment and providing real-time temperature data to the microcontroller. During testing, the DHT11 sensor consistently delivered stable and responsive readings, which were displayed on the LCD screen for user visibility.

To evaluate its performance, the sensor's output was compared against a calibrated reference thermometer under identical environmental conditions. The readings from the DHT11 were found to closely match those of the reference device, confirming the sensor's reliability and accuracy within an acceptable margin of error for this application.

Additionally, the system's ability to maintain consistent temperature readings over time was tested by observing the sensor response during prolonged operation and gradual temperature changes. The DHT11 demonstrated dependable performance, with minimal fluctuation or delay in response, ensuring that the fan speed was always adjusted appropriately based on the current room temperature.

Overall, the temperature sensing component of the system performed effectively, providing the foundation for the intelligent fan speed regulation and enhancing the overall efficiency of the temperature-controlled fan system.

4.1.2 BMS (Battery Management System)

The automatic temperature-controlled rechargeable DC fan system utilizes a Duravolt battery, managed by a Battery Management System (BMS) to ensure safe, efficient, and reliable operation. The BMS is a crucial component that monitors and protects the battery during charging and discharging processes.

One of its primary functions is to safeguard the battery from overcharging, over-discharging, and excessive current flow, which can otherwise lead to battery damage, reduced lifespan, or safety hazards. By keeping the duravolt battery operating within safe voltage and current limits, the BMS ensures the long-term reliability of the power supply.

4.1.3 LCD Display

The system utilizes an LCD display to present real-time information about the current room temperature and the DC fan speed level. The display serves as a user interface, allowing users to monitor system performance at a glance. The LCD was tested for accuracy by comparing the displayed temperature readings with actual measured values, and it consistently reflected correct and up-to-date data.

Additionally, the display's visibility and readability were evaluated under various lighting conditions, including both well-lit and dim environments. The LCD performed well in all scenarios, ensuring that users could clearly always view temperature values and fan speed levels. This reliability in visual feedback enhances user interaction with the system and contributes to the overall effectiveness of the temperature-controlled fan.

4.1.4 Speed Control

The system effectively and accurately controls the speed of the DC fan based on real-time temperature readings. Using Pulse Width Modulation (PWM), the fan's speed is automatically adjusted to match the current temperature increasing as the temperature rises and decreasing as it falls thereby providing efficient and responsive cooling.

During testing, the system met all key performance conditions, including quick response time, accurate speed adjustment, and operation within the specified speed range of the fan. It maintained a stable and consistent fan speed over time, without sudden fluctuations or delays, ensuring reliable performance throughout.

This dynamic speed control not only enhances user comfort by responding precisely to temperature changes but also contributes to energy efficiency and prolongs the life of the fan by preventing unnecessary high-speed operation when it's not needed.

4.2 Discussion

The results of testing the automatic temperature-controlled rechargeable DC fan system were highly encouraging. The system demonstrated its effectiveness in several critical areas, including accurate room temperature sensing, dynamic control of the DC fan speed, and clear visual feedback of both temperature and fan speed through the LCD display.

At the core of the system is the DHT11 temperature sensor, responsible for measuring ambient room temperature. This data is sent to the PIC microcontroller, which serves as the central control unit, making decisions based on temperature input. A Duravolt rechargeable battery powers the entire system, with a Battery Management System (BMS) in place to protect against overcharging, over-discharging, and current surges. The battery is charged using a 5V DC supply, allowing the system to operate even in the absence of mains power.

The LCD display provides real-time updates on both temperature and fan speed, while a status LED indicates the operational state of the system. The DC fan adjusts its speed automatically based on the detected temperature, ensuring a comfortable environment and promoting energy efficiency.

An added feature of the system is its voice command integration, which allows users to control the fan through spoken commands. The user can turn the fan ON or OFF, as well as increase or decrease the fan speed using simple voice inputs. This voice-controlled functionality improves accessibility, especially for elderly or physically challenged users, and adds a layer of convenience by enabling hands-free operation.

The overall design and construction of the system proved successful. The automated temperature-controlled rechargeable DC fan offers significant advantages, including energy efficiency, user comfort, and accurate, real-time temperature regulation. Its versatile design makes it suitable for a variety of applications where reliable cooling and environmental control are essential such as homes, offices, greenhouses, server rooms, and vehicles.

4.3 Presentation of Results

The behaviour of the automatic temperature-controlled DC fan system is largely determined by the data received from the DHT11 temperature sensor. In the absence of a signal, the regulator and sensors remain in their default programmed state, as no temperature data is received from the DHT11 sensor. Consequently, the microcontroller does not activate the fan since it is designed to respond only when specific temperature thresholds are met. Once a valid signal is detected by the DHT11 sensor, the microcontroller decodes the input and sends the appropriate command to control the DC fan based on the ambient temperature. This allows the fan to operate at varying speeds corresponding to temperature ranges, thereby ensuring adaptive environmental control.

As shown in Plate 4.1, when the ambient temperature is below 26°C, the system remains in a non-operational state, and the fan speed remains at 0. This confirms that the system conserves energy by only activating when necessary.

Upon reaching temperatures between 26°C and 31°C, the fan automatically transitions to speed 1, as illustrated in Plate 4.2, indicating that the system has registered a moderate rise in temperature and initiated airflow accordingly.

Further increases in temperature result in successive changes in fan speed. At temperatures ranging from 32°C to 37°C, the fan increases to speed 2, as seen in Plate 4.3. This is followed by speed 3 activation in the 38°C to 43°C range, demonstrated in Plate 4.4.

In Plate 4.5, the system's response to temperatures between 44°C and 49°C is depicted, with the fan operating at speed 4 to accommodate the elevated thermal environment. Finally, Plate 4.6 displays the system operating at its maximum setting speed 5 when temperatures exceed 50°C, highlighting the system's full functional range in response to extreme heat conditions.

These observations confirm that the fan system effectively adjusts its speed dynamically based on real-time environmental inputs, demonstrating its suitability for applications where automated thermal regulation is critical.



Plate 4.1 Automatic temperature-controlled DC fan at speed 0

The automatic temperature-controlled DC fan remains at a default speed of 0 (non-operational) when the ambient temperature is below the minimum threshold of 26°C.



Plate 4.2 Automatic Temperature-Controlled Fan: Speed 1 (26–31°C)

Automatic temperature-controlled DC fan = speed 1 (within the range of 26°C to 31°C (operational)).



Plate 4.3 Automatic Temperature-Controlled Fan: Speed 2 (32–37°C)

Automatic temperature-controlled DC fan = speed 2 (within the range of 32°C to 37°C (operational)).



Plate 4.4 Automatic Temperature-Controlled Fan: Speed 3 (38–43°C)

Automatic temperature-controlled DC fan = speed 3 (within the range of 38°C to 43°C (operational)).



Plate 4.5 Automatic Temperature-Controlled Fan: Speed 4 (44–49°C)

Automatic temperature-controlled DC fan = speed 4 (within the range of 44°C to 49°C (operational)).



Plate 4.6 Automatic Temperature-Controlled Fan: Speed 5 (50°C–∞)

Automatic temperature-controlled DC fan = speed 5 (within the range of 50°C to ∞ (operational)).

The voice-controlled functionality of the automatic temperature-controlled DC fan system adds a layer of interactive convenience and accessibility. In the absence of a voice input, the voice control module and microcontroller remain in their default programmed state, as no command is detected from the user. As a result, the system remains idle the microcontroller does not execute any control action, as it is programmed to respond only upon receiving a recognized and valid voice instruction.

As illustrated in Plate 4.7, when no voice command is detected, the fan remains at its default speed of 0, indicating that the system awaits user input. This non-responsive state ensures energy is conserved until user interaction prompts activation.

Once a recognized voice command is issued, for instance, to turn the fan ON or to increase its speed, the microcontroller processes the instruction and sends the appropriate signal to the DC fan. This enables responsive and adaptive control, particularly useful for hands-free operation.

Plate 4.8 clearly demonstrates this process: the system receives a valid voice command, decodes it, and increases the fan's speed to level 1. This successful transition from an idle to operational state reflects both the accuracy and responsiveness of the voice recognition system, validating its effectiveness in enhancing user experience and automation.

This voice-command capability ultimately supports improved accessibility for individuals with mobility challenges, while also promoting ease of use in modern smart home or workspace environments.



Plate 4.7 Automatic Temperature-Controlled Fan: Speed 0 (No Command Detected)

Plate 4.7 shows that the automatic temperature-controlled DC fan with voice command integration remains at a default speed of 0 because no command was detected.



Plate 4.8 Automatic Temperature-Controlled Fan: Speed 1 (Voice Command Activated)

Plate 4.8 illustrates the automatic temperature-controlled DC fan responding to a voice command by increasing its speed to level 1, indicating successful voice command initiation and system response

4.4 Contributions to Knowledge

This project contributes to the body of knowledge in the field of embedded systems, automation, and energy-efficient climate control by presenting a cost-effective and portable solution for automatic temperature regulation using a DC fan. The key contributions include:

1. **Integration of Temperature-Based Fan Speed Control:**

The project demonstrates the practical application of a temperature sensor (DHT11) in regulating the speed of a DC fan using Pulse Width Modulation (PWM), showcasing how environmental feedback can be used to drive real-time hardware responses for energy efficiency and user comfort.

2. **Use of PIC Microcontroller for System Automation:**

Unlike many similar projects that utilize Arduino platforms, this design uses a PIC microcontroller, broadening the application scope and encouraging the exploration of more advanced microcontroller architectures in embedded projects.

3. **Battery-Powered and Rechargeable Fan System:**

The incorporation of a rechargeable Duravolt lithium battery and a Battery Management System (BMS) offers a sustainable and off-grid solution for cooling, especially valuable in regions with unstable power supply. It enhances the understanding of battery safety, power management, and system autonomy.

4. **Voice Command Integration:**

The system introduces voice-controlled operation, allowing users to power the fan ON/OFF and adjust speed without manual input. This contributes to ongoing research in accessible and user-friendly interfaces, particularly for individuals with physical disabilities or mobility issues.

5. **System Simulation and Real-Life Implementation:**

By combining Proteus simulation with real-life circuit implementation, the project demonstrates the full development cycle, from design and testing to physical construction serving as a useful guide for future embedded system developers and researchers.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

The conclusion of this project was reached through a systematic process involving design, implementation, and performance evaluation of the automatic temperature-controlled rechargeable DC fan with voice command integration. The system was first developed using a circuit designed in Proteus, with key components like the PIC microcontroller, DHT11 sensor, LCD, and DC fan selected and tested individually for functionality. Pre-implementation testing verified the components using a multimeter, ensuring accurate voltage levels, continuity, and operational integrity before hardware assembly.

After successful coding and integration, the system was tested under varying temperature conditions to observe how well the fan responded through PWM-based speed control. Simultaneously, voice command functionality was implemented and tested using a predefined set of spoken commands, with results indicating real-time response to user input. The rechargeable battery setup was also examined for power stability and effectiveness during off-grid scenarios.

These stages, design, hardware testing, software integration, and field performance evaluation, collectively confirmed that the system met its objectives. It automatically adjusted fan speed in proportion to temperature, provided clear visual feedback on the LCD, and responded reliably to voice inputs. The system not only operated as expected but also demonstrated practical relevance and efficiency, leading to a confident conclusion about its usability and potential applications.

5.2 Limitations

While the automatic temperature-controlled rechargeable DC fan with voice command integration performed effectively, few limitations were identified during the design, construction, and testing phases of the project:

1. **Sensor Accuracy**

The DHT11 temperature sensor, while cost-effective and easy to implement, has limited precision and a slow response time. This occasionally caused delays or minor inaccuracies

in adjusting fan speed relative to actual temperature fluctuations, affecting real-time system responsiveness.

2. Limited Battery Capacity

The Duravolt rechargeable battery used in powering the system has restricted runtime, particularly when operating the fan at high speeds. This limits the system's ability to function continuously without frequent recharging, reducing its overall efficiency and portability.

3. Voice Command Limitations

Although voice command integration greatly improves user accessibility, its functionality was sometimes hindered by environmental noise, inconsistent pronunciation, and the need for specific, pre-programmed commands. These factors limited the flexibility and reliability of hands-free operation in varied conditions.

5.3 Recommendations

Based on the findings and limitations encountered during the design and implementation of the automatic temperature-controlled rechargeable DC fan with voice command integration, the following recommendations are proposed for future improvements and developments:

1. Upgrade to a More Accurate Temperature Sensor

Replacing the DHT11 with a more precise sensor such as the DHT22 or DS18B20 is essential to improve temperature accuracy and response time. This would ensure more reliable fan speed adjustments and overall system efficiency.

2. Use of a Higher Capacity Battery

Implementing a larger-capacity lithium-ion or lithium-polymer battery would significantly extend operational time, making the system more reliable and practical in areas with limited access to constant power supply.

3. Enhanced Voice Recognition Module

To increase usability and accessibility, integrating a more robust voice recognition system with better noise filtering and support for varied accents or offline operation would make voice commands more accurate and flexible, especially in noisy or low-connectivity environments.

REFERENCES

- Adeloye, M. Y., Kotoye, A. A., Idris, O. Y., and Egbedele, I. A. (2017). *Construction of a Temperature Control Fan Using a Microprocessor. American Journal of Engineering Research*, 6(9), 324–328.
- Amiri, N. (2007). *Voice-Activated Remote Control for Fan/Heater*.
- Anket, B., Sachin, C., Golsangi, S., Kushal, B., and Aparna, K. (2021). *Automatic Fan Speed Control System Using Microcontroller. Journal of Science and Technology*, 06(01).
- Arduino.cc. (2021). *Arduino Uno Rev3: Technical specifications, features, and applications*. Retrieved from <https://www.arduino.cc/en/Main/ArduinoBoardUno>
- Da Silveira, A. C., Rodrigues, E. C., Saleme, E. B., Covaci, A., Ghinea, G., and Santos, C. A. S. (2023). *Thermal and wind devices for multisensory human-computer interaction: an overview. Multimedia Tools and Applications*, 82(22), 34485–34512.
- Embedded Systems Blog. (2023). *Interfacing LCD Display with Arduino for Real-Time Monitoring*. Retrieved from <https://embeddedhub.com>
- Fan, J. R., Luo, Y. Q., Yang, J. H., and Lin, K. C. (2019). *Energy saving-motion activated smart fan design and implementation. International Journal of Engineering Creativity and Innovation*, 1, 24–32.
- HC-05 Bluetooth Module. (n.d.). *Bluetooth Module Specification and Applications*. Retrieved from <https://components101.com>
- Junizan, N. A., Asyikin, A. M., Nabilah, Z., and Norliza, I. (2019). *Design and implementation of automatic room temperature controlled fan using Arduino Uno and LM35 heat sensor. International Journal of Engineering Creativity and Innovation*, 1(2), 8–14.
- Khan, H. R., Ahmed, W., Masud, W., Alam, U., Arshad, K., Assaleh, K., and Qazi, S. A. (2024). *Design and Experimental Results of an AIoT-Enabled, Energy-Efficient Ceiling Fan System. Sustainability*, 16(12), 5047.

- Microchip Technology. (2021). *DHT11 Digital Temperature and Humidity Sensor Datasheet*. Retrieved from <https://datasheets.microchip.com/>
- Mustafa, S., Hossam, A., and Muammer, M. (2014). *Automatic Fan Speed Control System Using Microcontroller*. 6th Int'l Conference on Electrical, Electronics and Civil Engineering (ICEECE'2014), Nov. 27–28, 2014, Cape Town (South Africa).
- Nur, H., Shamsudin, A., Ibrahim, Z., and Rahim, M. (2018). *A Review on the Application of Rechargeable Batteries in HVAC Systems*. *International Journal of Power Electronics and Drive Systems*, 9(4), 1869–1879.
- Patil, B. N., Sonar, S., Karadiguddi, P., and Banti, S. (2016). *Speed Control Single Phase Induction Motor Using Android Phone*. *International Journal of Engineering and Techniques*, 2(3).
- Pimpalgaonkar, A., Jha, M., Shukla, N., and Asthana, K. (2013). *A precision temperature controller using an embedded system*. *International Journal of Scientific and Research Publications*, 3(12), 1–3.
- Sadiku, M. N. O., and Musa, S. M. (2020). *IoT Applications and Solutions*. *IEEE Access*, 8, 123456–123467.
- Senthilkumar, D. S., Brindha, D. K., and Kataria, S. H. (2017). *Temperature Control DC Fan*. 1st International Conference on New Paradigms in Engineering Technology and Management (pp. 416–424).
- Sharma, R. (2020). *Battery Management Systems for Safe and Reliable Energy Storage Solutions*. Retrieved from <https://energytechjournal.com>
- Shivshankar, A., Shivani, M., Patil, R., and Namrata, R. D. (2020). *Automatic Temperature Based Fan Speed Controller Using Arduino*. *International Journal of Advanced Research in Science and Technology*, 2(4).
- Wen, Y., and Jeng, M. (2008). *Diagnosable discrete event system design: A case study of an automatic temperature control system*. *IEEE International Conference on Systems, Man and Cybernetics*, 3691–3696.